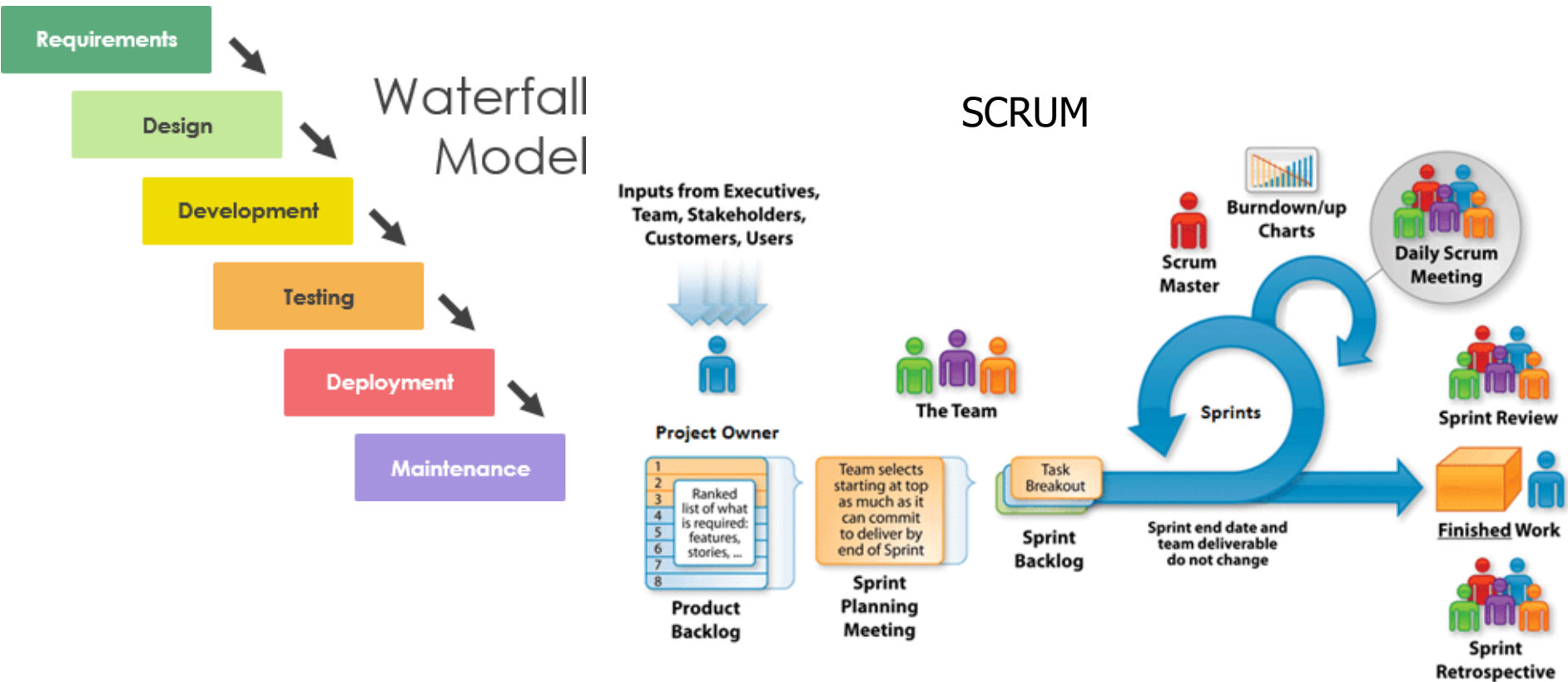


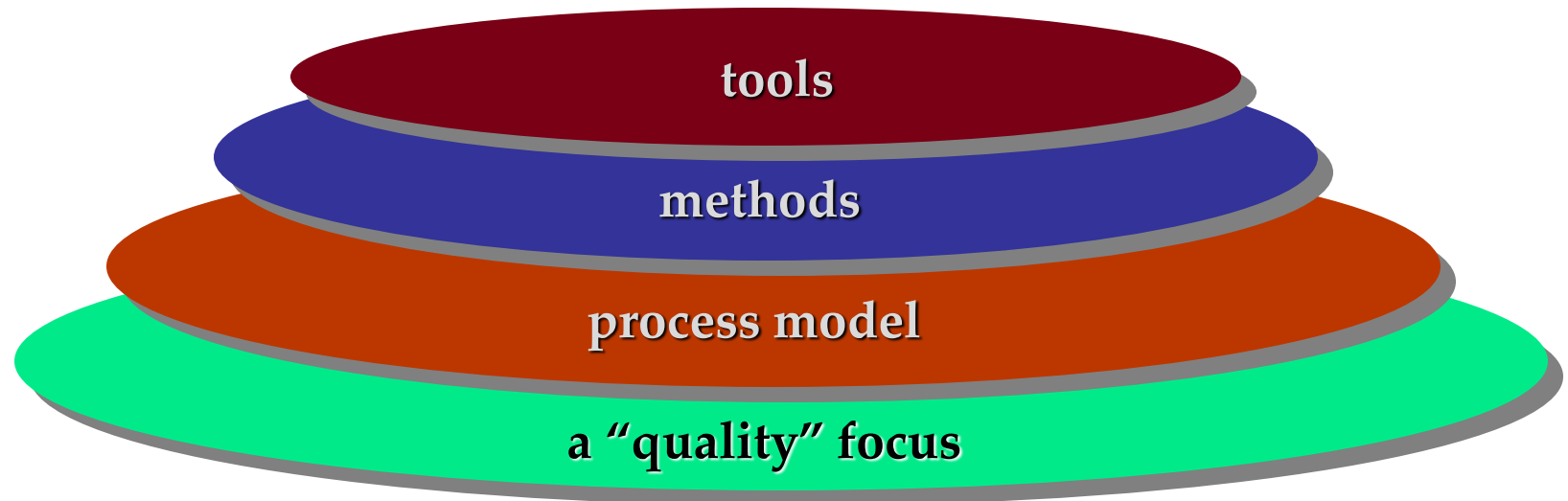
Process modeller med SCRUM

En Introduktion

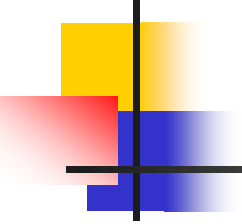


A Layered Technology

Which process model are part of



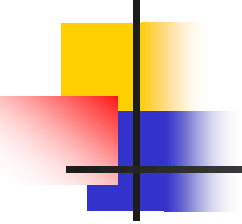
Software Engineering



Problems in Software development

in the early days

- **People** developing systems were **consistently wrong** in their **estimates** of **time, effort, and costs**
- **Reliability** and **maintainability** were **difficult** to achieve
- **Delivered** systems frequently did **not work**
 - **1979** study of a small number of **government projects** showed that:
 - 2% worked
 - 3% could work after some corrections
 - 45% delivered but never successfully used
 - 20% used but extensively reworked or abandoned
 - 30% paid and undelivered
- **Fixing bugs in delivered** software **produced more bugs**
- Etc...



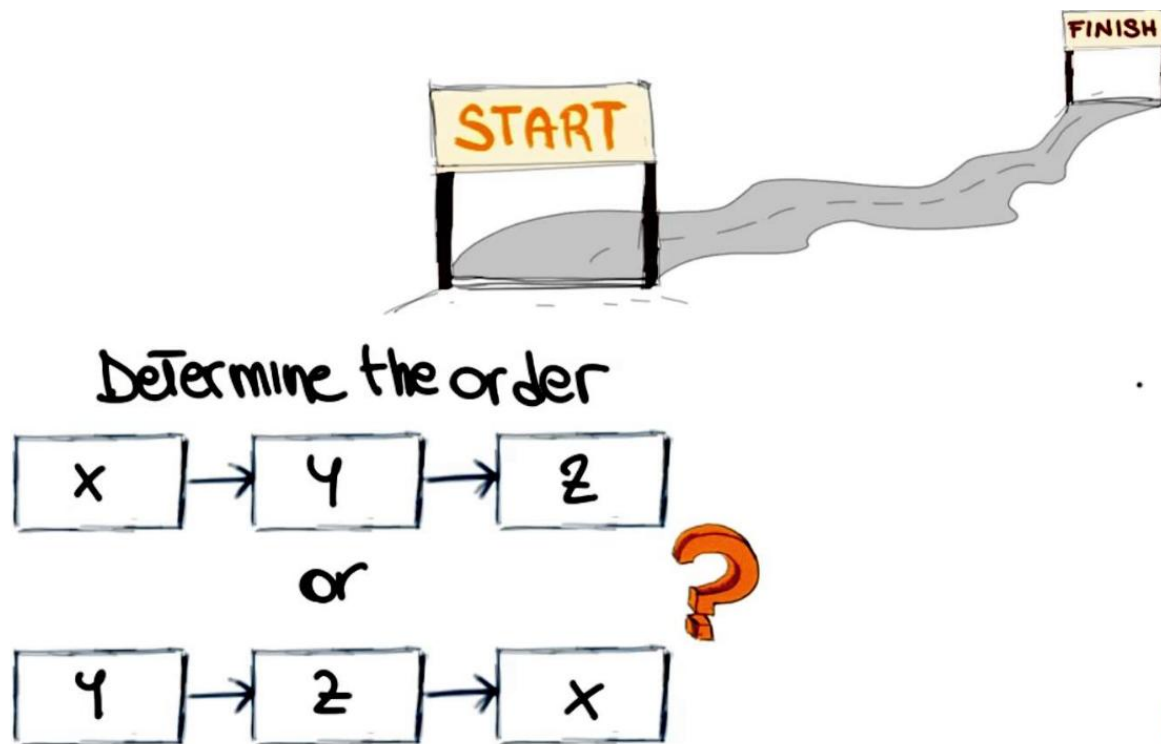
Software Lifecycle

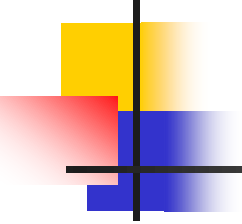
Steps or phases in a **software lifecycle** fall generally into these categories:

- **Requirements** (Relative Cost 2%)
- **Specification** (analysis) (Relative Cost 5%)
- **Design** (Relative Cost 6%)
- **Implementation** (Relative Cost 5%)
- **Testing** (Relative Cost 7%)
- **Integration** (Relative Cost 8%)
- **Maintenance** (Relative Cost 67%)
- **Retirement**

What is a Software Process Model?

A software Process Model is a description of the sequence of activities carried out in a software engineering project and the relative order of these activities.

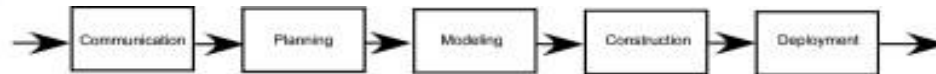




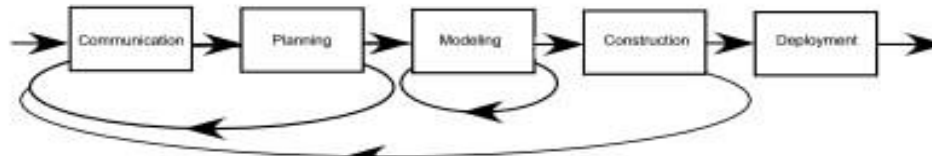
Software Process Models

- Software process models are general approaches for organizing a project into activities
 - Help the project manager and the team to decide:
 - **What work should be done?**
 - **In what sequence to perform the work?**
 - **Important!** The models should be seen as *aids to thinking*, not rigid prescriptions of the way to do things

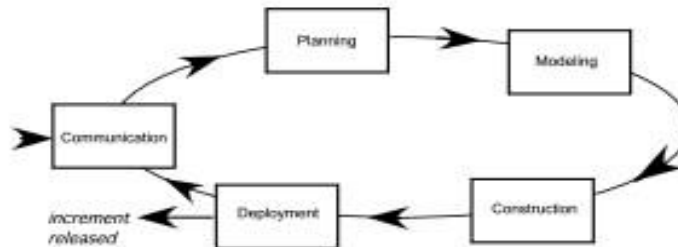
Process Flow



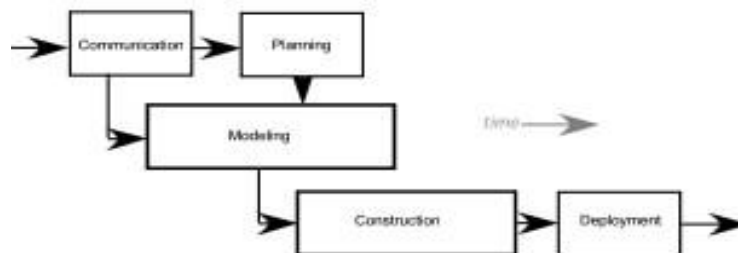
(a) linear process flow



(b) iterative process flow

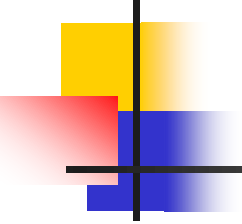


(c) evolutionary process flow



(d) parallel process flow

The process flow describes how the framework activities are organized with respect to sequence and time



Prescriptive Models

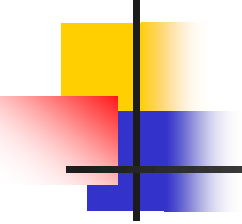
- Prescriptive process models advocate an orderly approach to software engineering
- ...
- Prescriptive process models strive for structure and order.

The Waterfall Model

The incremental Model

Evolutionary Models

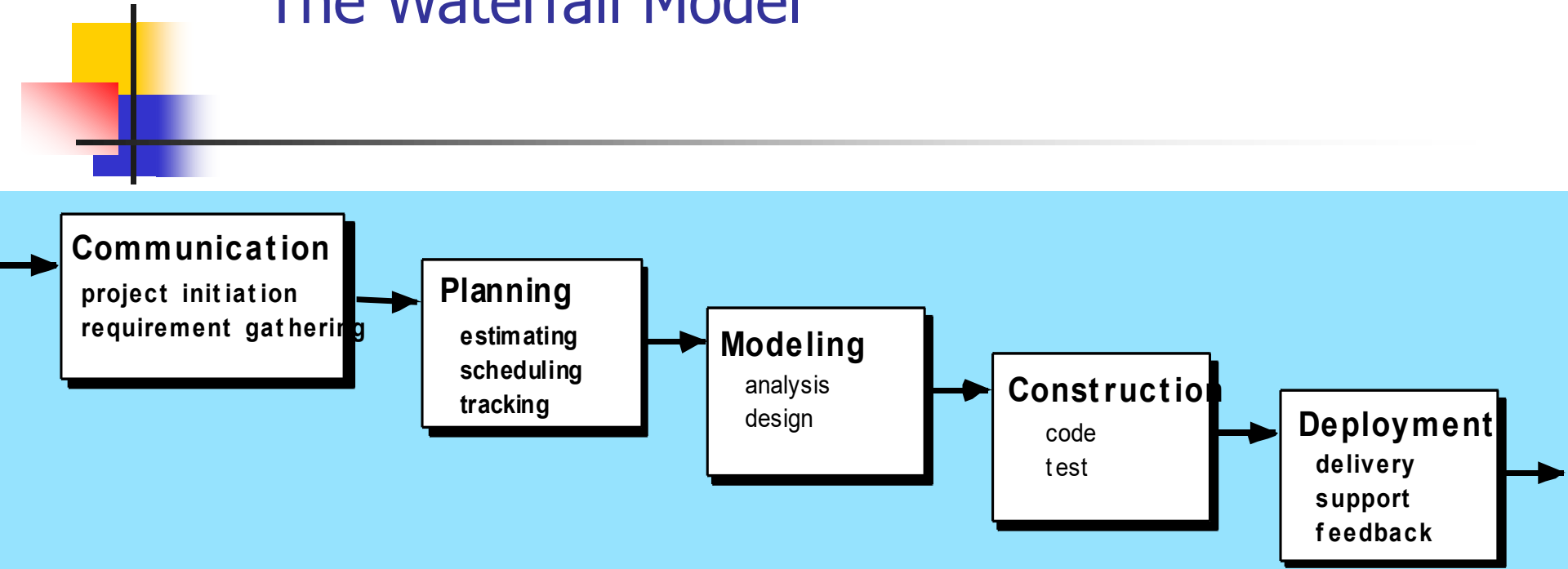
- **Prototyping**
- **Spiral Model**



Waterfall model

- A term coined in about 1969
- An effort to **move** the development of software
 - **Magic** - which only a select few can do
 - **Art** - which the talented can do
 - **Science** - which supposedly anyone can do!

The Waterfall Model



Linear process flow

Discuss following questions with the student sitting next to you:

- Drawbacks and advantages of this process model?
- When is the waterfall model appropriate?

Critique of Waterfall Model



- The model implies that you should attempt to complete a given stage before moving on to the next stage
 - Does not account for the fact that requirements constantly change.
 - It also means that customers cannot use anything until the entire system is complete.
- Assumes understanding of problem and full requirements early on
- It implies that you can get the requirements right by simply writing them down and reviewing them.

Critique of Waterfall Model cont.



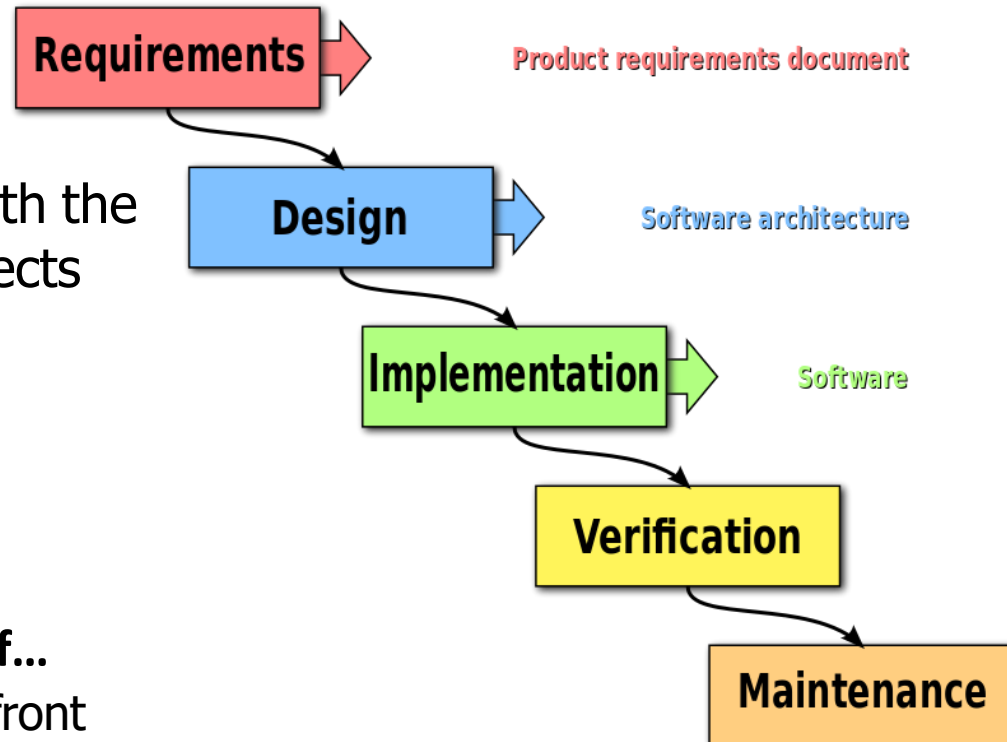
- The model implies that once the product is finished, everything else is maintenance.
- Assumes patience from customer
- Surprises at the end are very expensive

This model is only appropriate when the requirements are well-understood and changes will be fairly limited during the design process.

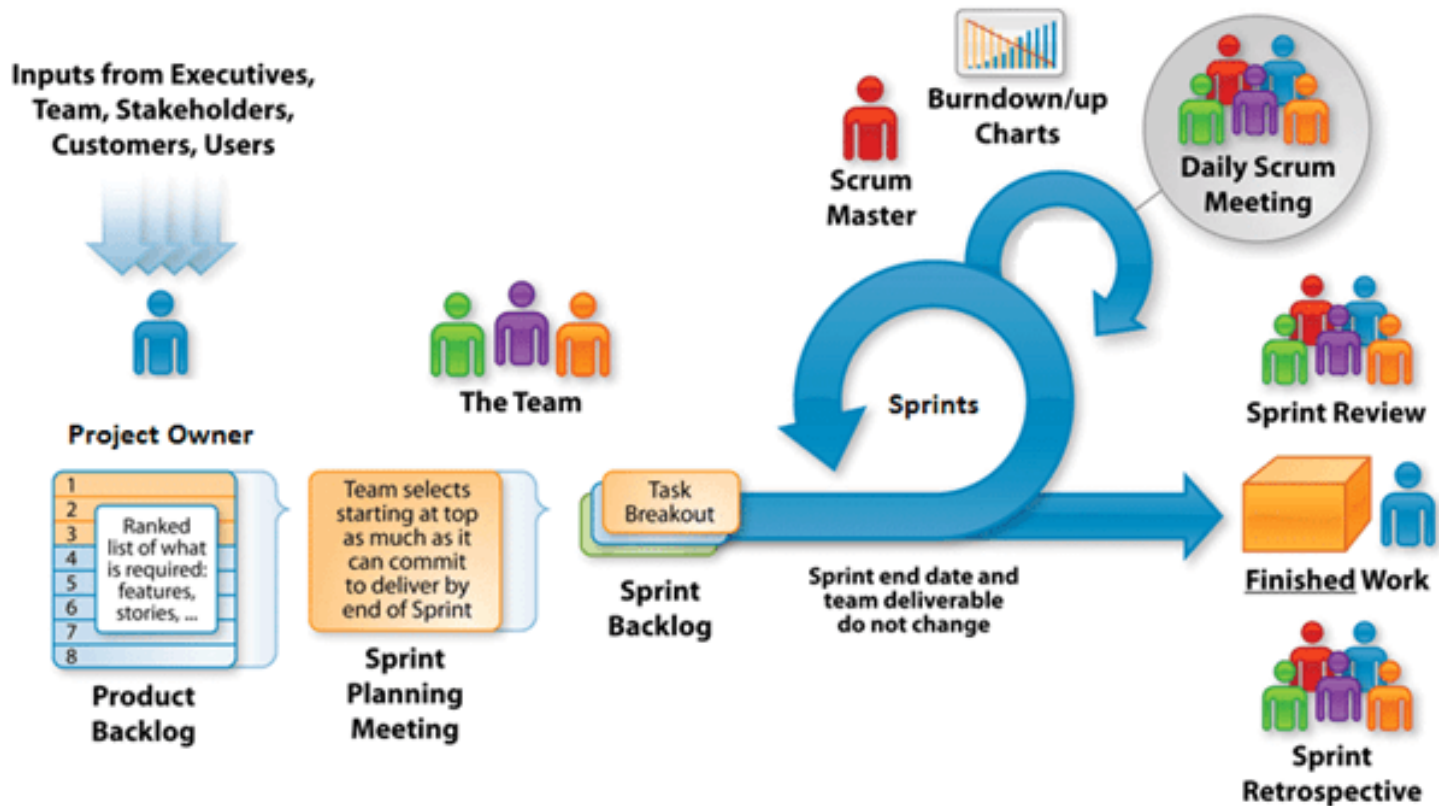
Waterfall

- Waterfall is strongly associated with the highest failure rates for software projects

- But, it's still a “very good process” if...
 - You know all the requirements upfront
 - ...and the requirements doesn't change



SCRUM



**** Watch the video: SCRUM in under 10 minutes ****

SCRUM

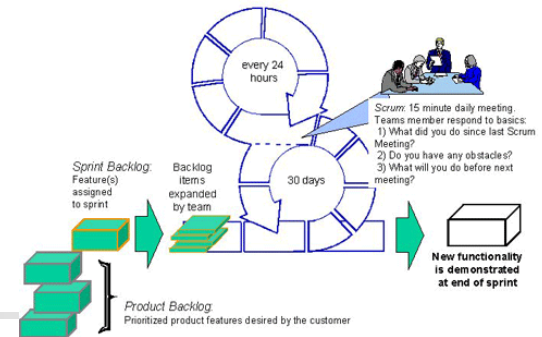
Roles



- The **Product Owner** represents the voice of the customer. The Product Owner writes customer-centric items, prioritizes them and then places them in the *product backlog*.
- The **ScrumMasters** job is to remove impediments to the ability of the team to deliver the sprint goal. The ScrumMaster is not the leader of the team (as the team is self-organizing) but acts as a buffer between the team and any distracting influences. The ScrumMaster is the enforcer of rules.
- **The team** has the responsibility to deliver the product. A team is typically made up of 5–9 people with cross-functional skills who do the actual work (design, develop, test, technical communication, etc.). Scrum enables the creation of self-organizing teams by encouraging co-location of all team members

SCRUM

Process

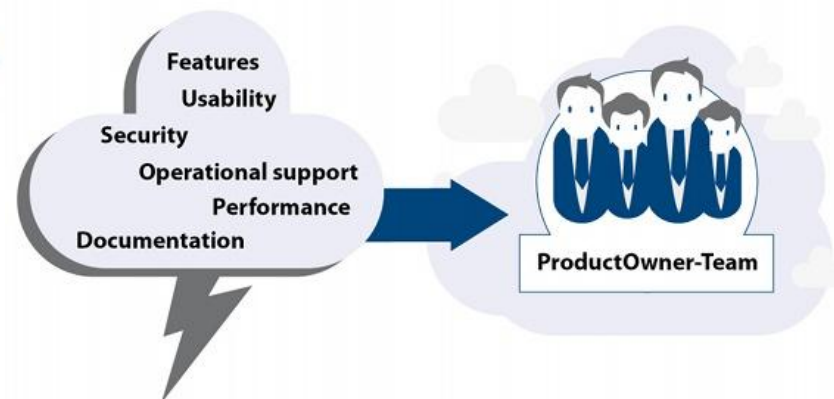


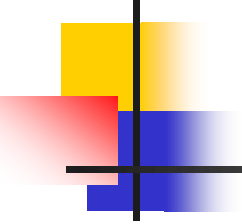
- During each **Sprint**, the team creates a potentially shippable product increment (e.g.working and tested software).
- The set of features that go into a sprint come from the **product backlog**, which is a prioritized set of high-level requirements.
- Which backlog items go into the sprint is determined during the **sprint planning meeting** (XP: planning game). The Product Owner informs the team of the items in the product backlog that he or she wants completed. The team then determines how much of this they can commit to complete during the next sprint.

The Product Backlog

The Product Backlog lists all features, functions, requirements, enhancements, and fixes that constitute the changes to be made to the product in future releases.

The Product Backlog is an ordered list of everything that is known to be needed in the product.





Waterfall vs Scrum

Waterfall

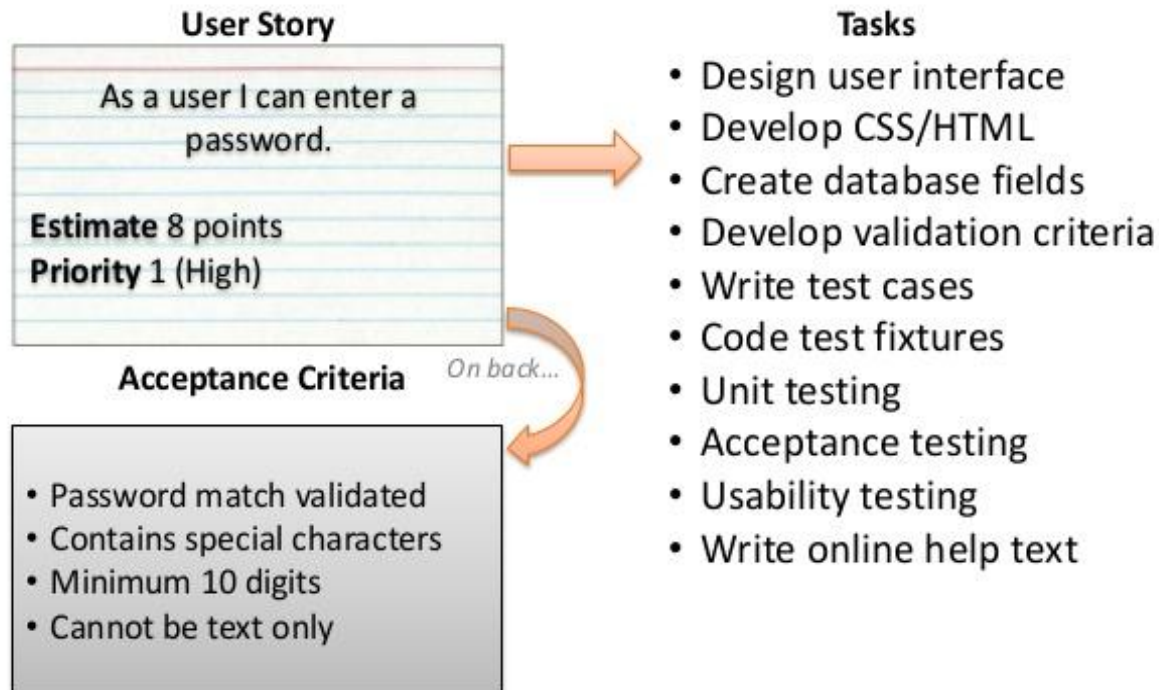
- Lineær, sekventiel proces
- Krav fastlægges på forhånd
- Kunden ser produktet sent
- God til stabile domæner

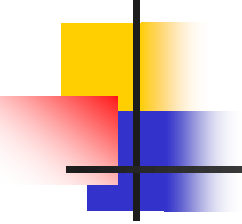
Scrum

- Iterativ og inkrementel
- Krav udvikles løbende
- Kunden ser produktet ofte
- God til dynamiske domæner

Decomposition of User Stories

From User Stories to Tasks





Estimation

Story Point based on:

- The amount of work to do
- The complexity of the work
- Any risk or uncertainty in doing the work

The main benefit of story points is that they allow individuals with different skills, experience levels and backgrounds to agree on a relative estimate for work.

Planning Poker:

The development team conducts an estimation, using the Fibonacci number (1, 2, 3, 5, 8, 13, 21...) to estimate the story point.

When estimating new stories all you have to do is pick a story and say: “will this take longer than reference story.”

Estimation

Three point value

BUT: When you have to define the tasks for a story, you often estimate in person-hours. Person hours must be corrected for personal time; experience & skills; use of tools etc.. (also called productivity factor – ex: estimate=12 h./factor 0,8 = 15h as duration)
Use Three Point Estimation – because you find the weighted average as a likely duration, and you find the standard deviations (the uncertainty in your estimate)

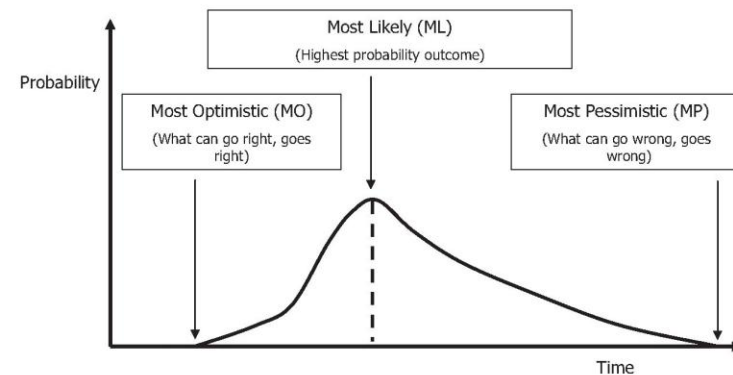
Formula

Example

$$\text{Duration (M)} = \frac{A + 3B + C}{5} = (5 + 3 \cdot 7 + 8) / 5 = 6,8$$

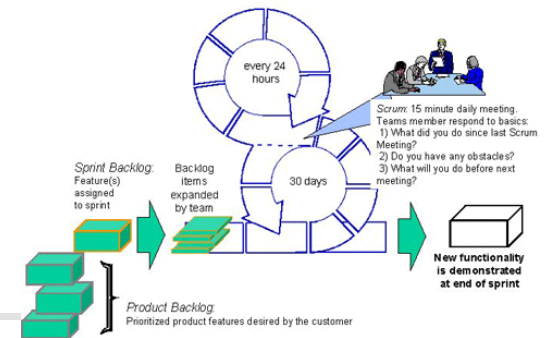
$$\text{Standard dev.} = \frac{C - A}{5} = (8 - 5) / 5 = 0,6$$

A= Optimistic; B=Most Likely; C = Pessimistic

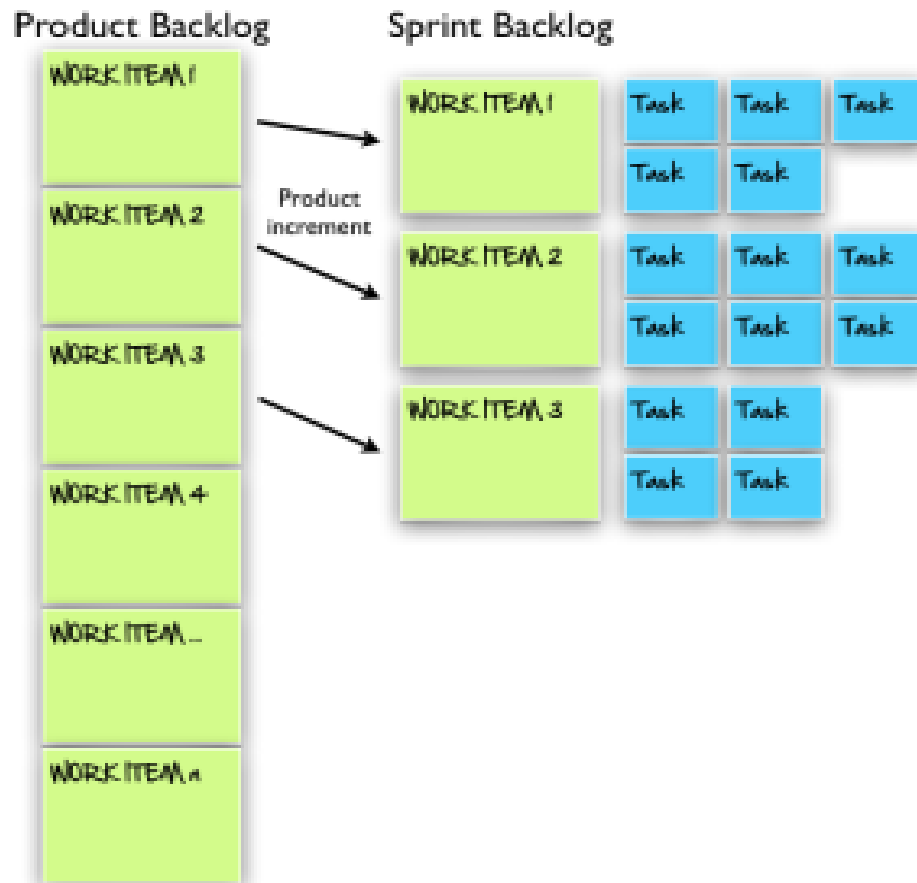


SCRUM

Process

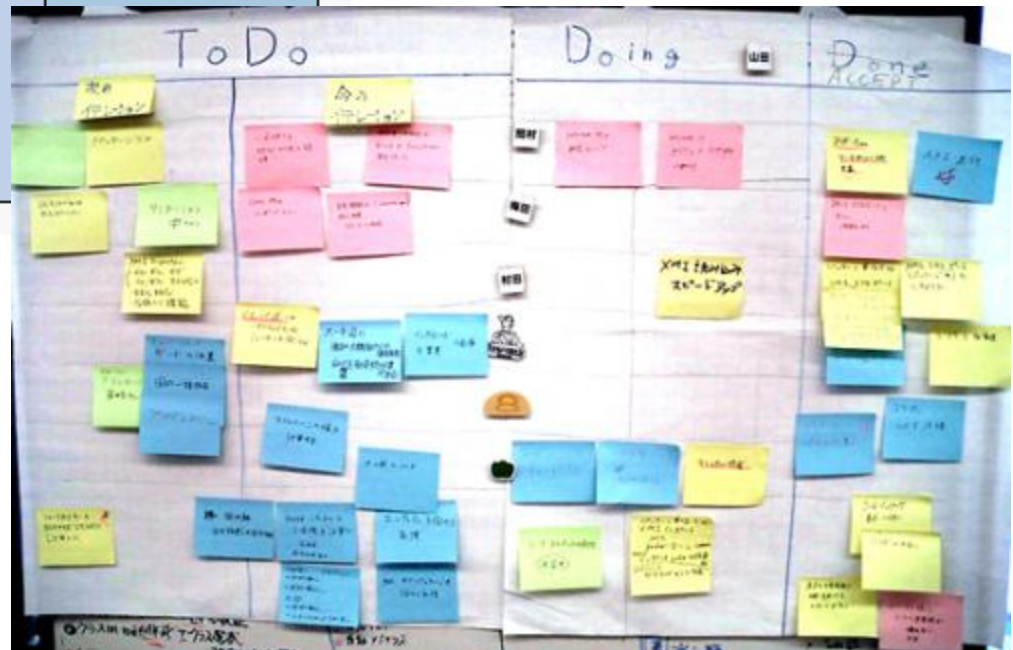


- During a sprint, no one is allowed to change the sprint backlog, which means that the requirements are **frozen** for that sprint.



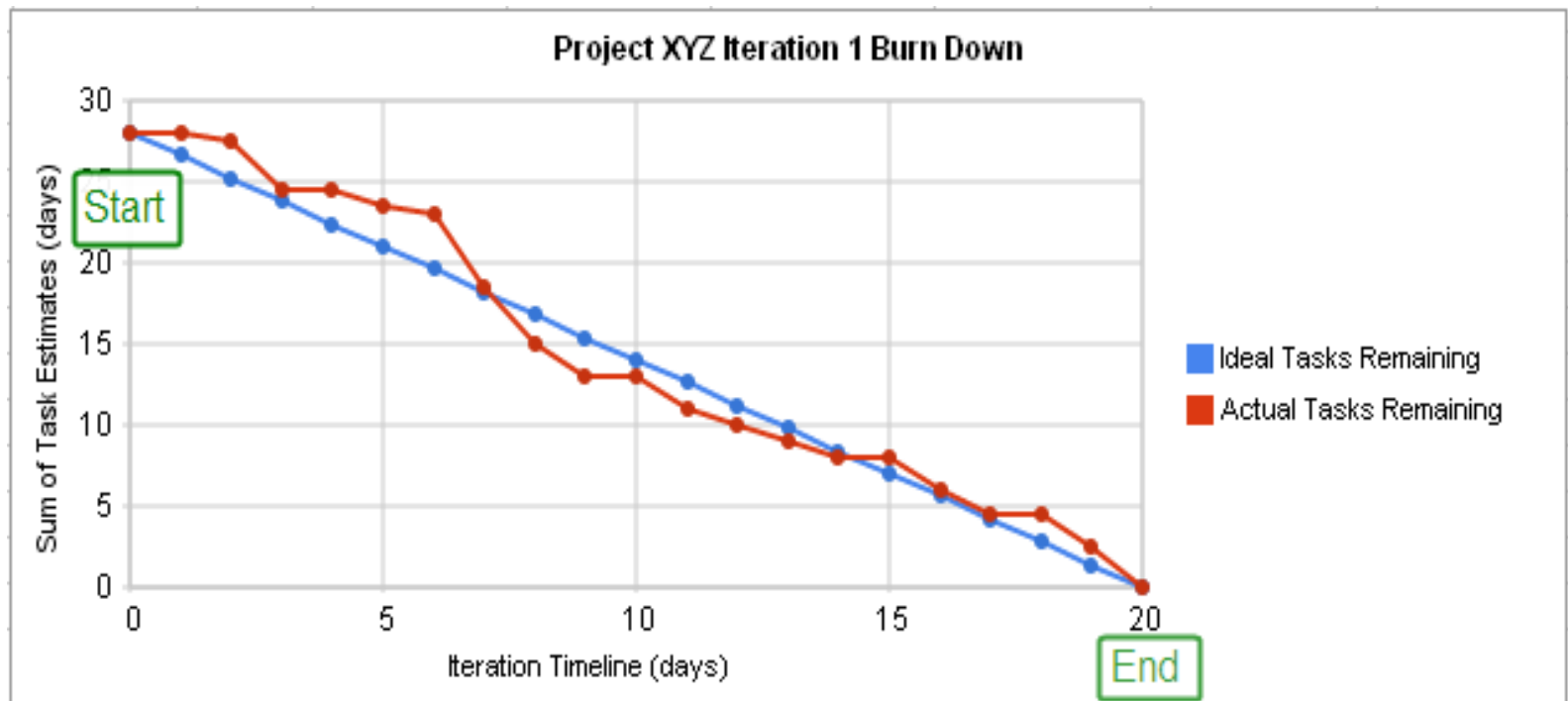
Scrum Board

Scrum Board			
Story	To Do	In Progress	Done
Story	Task 3 Task 4	Task 2	Task 1



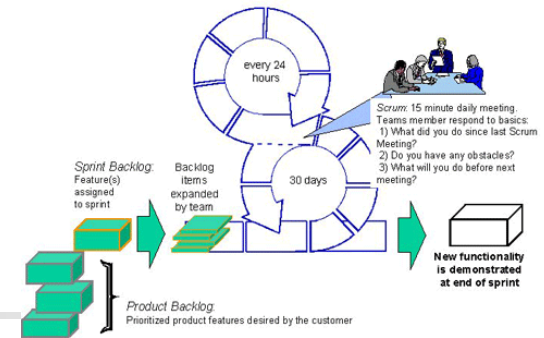
The Burn Down Chart

- A **burn down chart** is a graphical representation of work left to do versus time. To motivate the team, the sprint burndown chart should be displayed prominently.



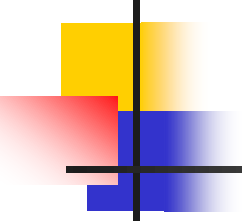
SCRUM

Process



- Each day during the sprint, a project status meeting occurs -a **"daily scrum"**.
- The meeting is timeboxed to 15 minutes.
- Each team member answers three questions:
 - What have you done since yesterday?
 - What are you planning to do today?
 - Do you have any problems preventing you from accomplishing your goal





Other meetings

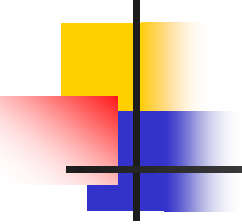
At the end of a sprint cycle, two meetings are held: the "**Sprint Review Meeting**" and the "**Sprint Retrospective**".

At the Sprint Review Meeting:

- Review the work that was completed and the planned work that was not completed
- Present the completed work to the stakeholders (a.k.a. "the demo")
- Incomplete work cannot be demonstrated

At the Sprint Retrospective:

- All team members reflect on the past sprint
- Make continuous process improvements
- Two main questions are asked in the sprint retrospective: What went well during the sprint? What could be improved in the next sprint?



Opsummering

- 📖 Udvikling organiseres med procesmodeller
- ⬇️ Vandfaldsmodellen: sekventiel, god ved stabile krav
- 🔄 Scrum: iterativ, fleksibel, kundedrevet
- 👥 Scrum bruges via roller, backlog, sprint og møder